Principal component analysis (PCA) was performed on the gene expression dataset containing measurements for the 156 most differentially expressed genes across T cell samples. PCA reduces the dimensionality of the dataset while preserving the major sources of variation in gene expression. The scree plot indicated that the first few principal components explain a substantial proportion of the total variance, suggesting that the high-dimensional dataset can be effectively summarized using a small number of components.

Visualization of the samples in the PC1–PC2 space revealed clear clustering according to T cell type. Naïve (NAI), effector (EFFE), and memory (MEM) T cells form distinct groups, indicating systematic differences in their gene expression profiles. Memory T cells appear positioned between naïve and effector cells along the principal component axes, suggesting that their gene expression patterns are intermediate between these two states. This observation is consistent with findings reported by Holmes et al. (2005), which proposed that memory T cells exhibit transcriptional profiles between naïve and effector T cells.

The PCA loading plot of genes highlights those genes contributing most strongly to the principal components. Genes located farther from the origin have larger loadings and therefore play a greater role in driving the observed variation among samples. These genes likely contribute to the biological differences between the T cell states.